

Introduction

Definition:

The lipids are a heterogeneous group of compounds, which are related more by their physical than by their chemical properties. Despite lipids are used by humans since ancient times, they were studied only through the nineteenth century by scientists probably because they are difficult to separate and identify. No precise definition of the term lipid exists. In its broadest sense, "lipid" defines substances including fats, oils, steroids, waxes, and related compounds, which can be only characterized by a large array of properties.

They are in general:

Insoluble or immiscible with water but soluble in organic solvents such as chloroform, ether, benzene, acetone.

Lipids were also defined as natural substances having in their molecular structure one chain of at least 10 carbon atoms.

Formed of long-chain hydrocarbon groups (carbon and hydrogen) but may also contain oxygen, phosphorus, nitrogen and sulfur.

Coming from animals and plants living or fossilized, but they can also be produced by chemical synthesis.

Lipids are fatty acids and their derivatives, and substances related biosynthetically or functionally to these compounds.

Biological Functions:

❖ Energy storage:

Lipids are one of the three main classes of food, besides carbohydrates and proteins. They are responsible for storing energy that animals do not immediately need.

Important dietary constituents:

Not only because of their high energy value but also because of the fat-soluble vitamins and the essential fatty acids contained in the fat of natural foods.

❖ Structure of cell membranes:

Lipids are major components of cell membranes, and are responsible for most of the permeability filter functions of membranes. Membranes act as barriers to separate compartments within eukaryotic cells, and to separate all cells from their

surroundings. Now, every individual lipid class has been found to have some unique biological role that is distinct from its function as a source of energy or as a simple construction unit of a membrane.

Non-polar lipids act as electrical insulators, allowing rapid propagation of depolarization waves along myelinated nerves.

Precursors of hormones (steroids and prostaglandins) and vitamins.

Thermal insulator in the subcutaneous tissues and around certain organs.

In small quantities also as: cofactors, electron carriers, Signaling molecules (chemical signaling of various kinds) and cellular messengers.

Protective coatings of plants and insects.

Knowledge of lipid biochemistry is necessary in understanding many important biomedical areas, eg, obesity, diabetes mellitus, atherosclerosis, and the role of various polyunsaturated fatty acids in nutrition and health.

Classifications:

Lipids Covers a wide range of substances which contain moieties belonging chemically to hydrocarbons, alcohols, aldehydes, acids, or amines. The majority forms molecules such as waxes, triacylglycerols, phospholipids and glycolipids. Other substances are also considered as lipids while belonging to other classes of molecules, e.g. fat soluble vitamins (A, D, E and K), coenzymes (ubiquinones), pigments (carotenoids), terpenes, phenolics and sterols.

❖ **According to functions:**

Storage lipids: (Triglycerides)

Structural lipids: (phospholipids, cholesterol)

Functional lipids: (most steroids, glycolipids)

❖ **According to structure:**

The first group has open chains (saponifiable):

This group consists of lipids with a head group (largely polar) and nonpolar tails of varying size. This group includes fatty acids, triglycerides (triacylglycerols), waxes, phospholipids, sphingolipids, and glycolipids.

The nonpolar tails make these lipids hydrophobic.

The second group has closed chains (non-saponifiable):

The second group has fused-rings. This group includes cholesterol, steroid hormones and derived lipids.

❖ According to simple and complex:**Simple lipids:**

Esters of fatty acids with various alcohols.

Fats/Oils: Esters of fatty acids with glycerol. Fats are in the solid state where Oils are fats in the liquid state.

Waxes: Esters of fatty acids with higher molecular weight monohydric alcohols.

Complex lipids:

Esters of fatty acids containing groups in addition to an alcohol and a fatty acid.

Phospholipids: Lipids containing, in addition to fatty acids and an alcohol, a phosphoric acid residue. They frequently have nitrogen containing bases and other substituents, eg, glycerophospholipids the alcohol is glycerol and in sphingophospholipids the alcohol is sphingosine.

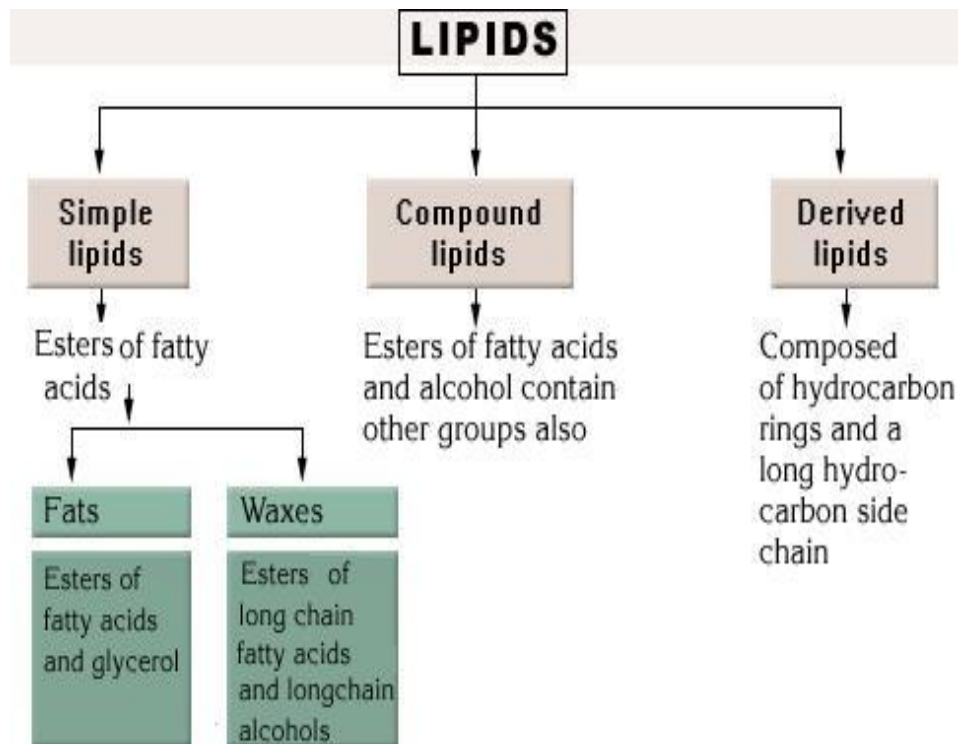
Glycolipids (glycosphingolipids): Lipids containing a fatty acid, sphingosine, and carbohydrate.

Other complex lipids such as sulfolipids and aminolipids. Lipoproteins may also be placed in this category.

Precursor and derived lipids:

These include fatty acids, glycerol, steroids, other alcohols, fatty aldehydes, and ketone bodies, hydrocarbons, lipid-soluble vitamins, and hormones.

Because they are uncharged glycerides cholesterol, and cholesteryl esters are termed **neutral lipids**.



Chemistry and Functions

Fatty Acids:

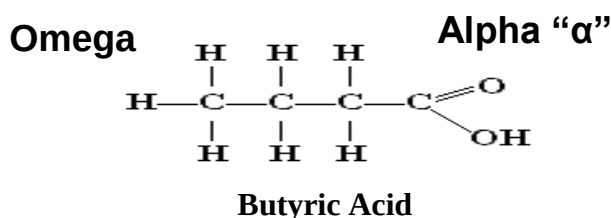
Definition:

Fatty acids are long-chain hydrocarbons with even numbers of carbons containing a carboxylic acid moiety at one end.

The numbering of carbons begins with the carbon of the carboxyl group which is readily ionized at physiological pH, rendering a negative charge onto fatty acids.

Chemical Structure of Fatty Acids:

Fatty acid have a -COOH group at the beginning of the chain, (the "alpha" end). The opposite end is called the omega.



Carbon atoms are numbered from the carboxyl carbon (carbon No. 1). The carbon atoms adjacent to the carboxyl carbon (Nos. 2, 3, and 4) are also known as the α , β , and γ carbons, respectively, and the terminal methyl carbon is known as the ω or n-carbon.

Nomenclature:

Somewhat confusing, The nomenclature designations used for fatty acids come from common name, fatty acid name, and the number of carbon atoms.

eg. Stearate – stearic acid – C18:0 – n-octadecanoic acid.

The most frequently used systematic nomenclature names the fatty acid after the hydrocarbon with the same number and arrangement of carbon atoms, with -oic being substituted for the final -e (Genevan system). Thus, saturated acids end in -anoic, eg, octanoic acid, and unsaturated acids with double bonds end in -enoic, eg, octadecenoic acid (oleic acid). The numeric designations used for fatty acids come from the number of carbon atoms, followed by the number of sites of unsaturation (eg, palmitic acid is a 16-carbon fatty acid is designated by (16:0). The site of unsaturation is indicated by the symbol Δ (delta) and the number of the first carbon of

the double bond (e.g. palmitoleic acid is a 16-carbon fatty acid with one double bond between carbons 9-10, is designated by 16:1 Δ^9).

Various conventions use Δ for indicating the number and position of the double bonds eg, Δ^9 indicates a double bond between carbons 9 and 10 of the fatty acid; ω 9 indicates a double bond on the ninth carbon counting from the ω - carbon. In animals, additional double bonds are introduced only between the existing double bond (eg, ω 9, ω 6, or ω 3) and the carboxyl carbon, leading to three series of fatty acids known as the ω 9, ω 6, and ω 3 families, respectively.

Numerical Symbol	Common Name	Structure	Comments
14:0	Myristic acid	$\text{CH}_3(\text{CH}_2)_{12}\text{COOH}$	Found attached to the N-term. of plasma membrane-associated cytoplasmic proteins
16:0	Palmitic acid	$\text{CH}_3(\text{CH}_2)_{14}\text{COOH}$	End product of mammalian fatty acid synthesis
16:1 Δ^9	Palmitoleic acid	$\text{CH}_3(\text{CH}_2)_5\text{C}=\text{C}(\text{CH}_2)_7\text{COOH}$	
18:0	Stearic acid	$\text{CH}_3(\text{CH}_2)_{16}\text{COOH}$	
18:1 Δ^9	Oleic acid	$\text{CH}_3(\text{CH}_2)_7\text{C}=\text{C}(\text{CH}_2)_7\text{COOH}$	
18:2 $\Delta^9,12$	Linoleic acid	$\text{CH}_3(\text{CH}_2)_4\text{C}=\text{CCH}_2\text{C}=\text{C}(\text{CH}_2)_7\text{COOH}$	Essential fatty acid

18:3 ^{Δ9,12,15}	Linolenic acid	$\text{CH}_3\text{CH}_2\text{C}=\text{CCH}_2\text{C}=\text{CCH}_2\text{C}=\text{C}(\text{CH}_2)_7\text{COOH}$	Essential fatty acid
20:4 ^{Δ5,8,11,14}	Arachidonic acid	$\text{CH}_3(\text{CH}_2)_3(\text{CH}_2\text{C}=\text{C})_4(\text{CH}_2)_3\text{COOH}$	Precursor for <u>eicosanoid synthesis</u>

Classification of Fatty Acids:

Fatty acid classification is based on the number of double bonds.

❖ Saturated fatty acids:

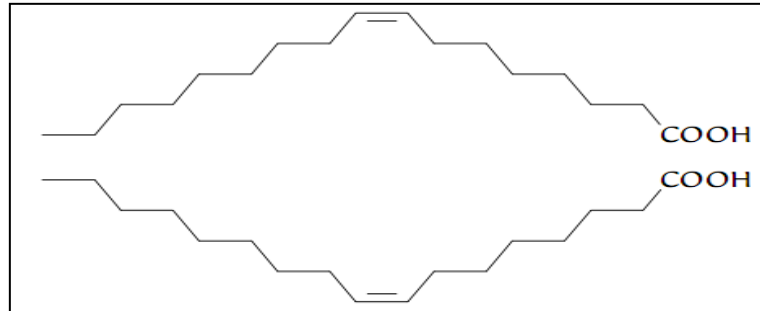
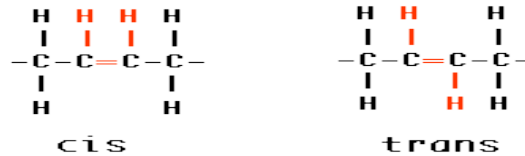
They have no double bonds between carbons. straight chains align well & pack closely in the solid state. have high melting point temperatures and are solids at room temperature. Saturated fats are found in animal products such as butter, cheese, whole milk, ice cream, and fatty meats. They are also found in some vegetable oils-coconut, palm and palm kernel oils.



Harmful Effects: Saturated fatty acids are a major risk factor for heart attacks and strokes. Diets high in saturated fatty acids increase the production of acetate fragments in the body which, in turn, leads to an increase in the production of cholesterol. When consumed, saturated fats tend to clump together and form deposits in the body, along with protein and cholesterol. They get lodged in blood cells and organs, leading to many health problems, including obesity, heart diseases, and cancers of the breast and colon. The build-up in the arteries can cause a narrowing of the arteries, called atherosclerosis, which consequently can lead to major heart problems.

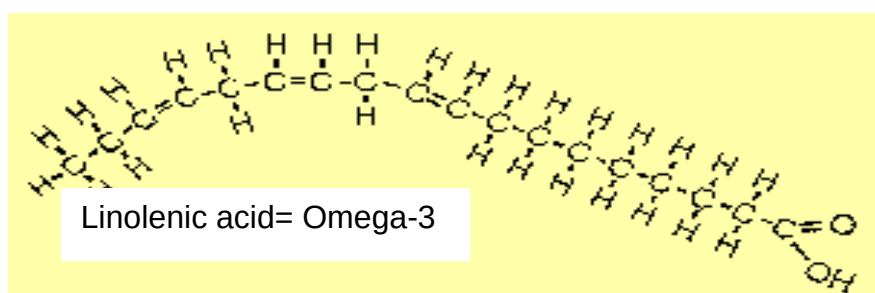
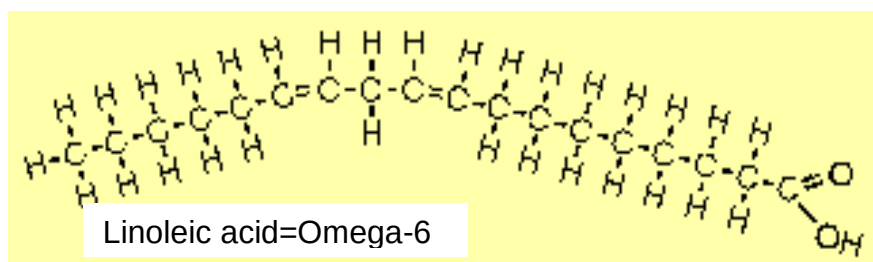
❖ Unsaturated fatty acids:

They have one or more double bonds. Most unsaturated fatty acids have cis double bonds which creates “kinks” in chains that do not align well and cannot pack closely in the solid state. have low melting point temperatures and are liquids at room temperature.



Unsaturated fats help to lower blood cholesterol if used in place of saturated fats and have a lot of calories, so you still need to limit them. There are two types: mono-unsaturated and polyunsaturated. Most liquid vegetable oils are unsaturated with the exceptions of coconut, palm, and palm kernel oils.

Omega-3,-6 and-9: The acid end is called the "alpha", and its opposite is the "omega" end. The first double-bond kink in an omega-3 fatty acid occurs between carbons 3 and 4 away from the omega end. For an omega-6, between carbons 6 and 7. And for omega-9, between carbons 9 and 10. Oleic acid is an omega-9 fatty acid.



These two fatty acids-alpha-linolenic acid (ALA, an omega-3) and linoleic acid (an omega-6)-are the two essential fatty acids that the human body needs and cannot manufacture.

benefits of omega fats in the diet:

Lower blood cholesterol levels, reducing a risk factor in coronary heart disease

Improve blood vessel elasticity .

Make the blood less sticky and less likely to clot.

Reduce inflammation and boost the immune system.

Contribute to the normal development of the foetal brain.

Trans fatty acids:

Trans fatty acids are rare in nature. They are only created in the rumen of cows and sheep, and are only found in small amounts in milk, cheese, beef and lamb. Trans fatty acids are also created during the manufactures like some table margarines and solid lipids used in the food industry.

Food manufacturers discovered that addition of hydrogen through polyunsaturated oils created "partially hydrogenated" fats that were less able to become rancid than the original oils and therefore had a longer shelf life.

Trans fatty acids have been linked to increased insulin resistance and type 2 diabetes as well as heart diseases. They appear to be worse for our health than saturated fats, largely because the body tries to use them in ways that evolution designed for unsaturated fats. Consumption of trans fats also raises LDL cholesterol and lowers HDL-the exact opposite of what one wants.

Eicosanoids

Including prostaglandins, thromboxanes, and leukotrienes which are derived from arachidonic acid and acts as paracrine hormones.

Biological Activity of Eicosanoids:

Prostaglandins & their related compounds initiate & regulate most physiologic functions:

Systems affected: nervous, smooth muscle, blood, reproductive.

Functions regulated: blood pressure, clotting & thinning, also, digestion, and breathing.

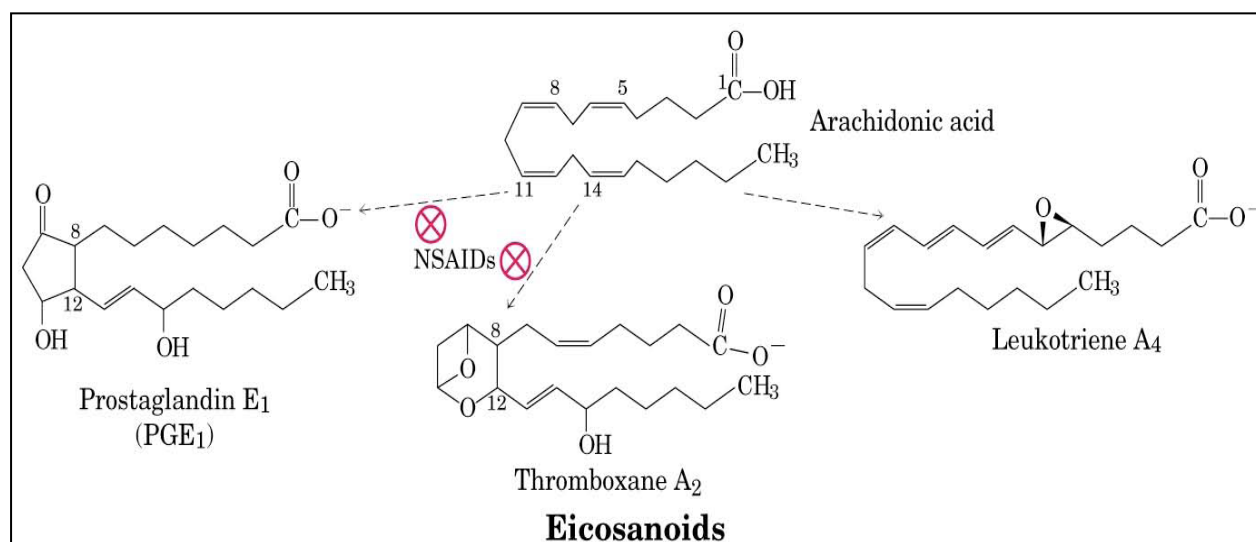
Responses initiated: pain, inflammation, fever.

They are often complementary in their biological activities “off/on switches”

Thromboxanes: platelet aggregators (blood clotting)

Prostacyclins: inhibit platelet aggregators (blood thinning)

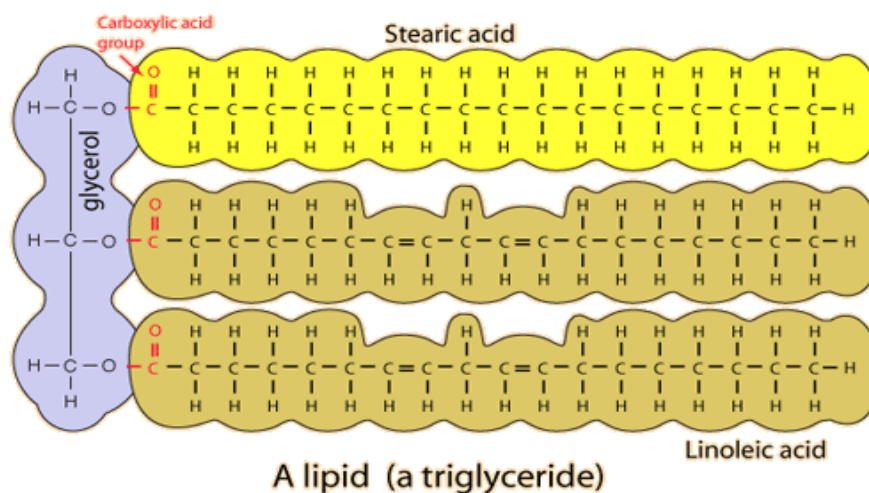
Eicosanoids, their inhibitors and their synthetic derivatives are widely used as medicinals



Simple lipids

Fats and oils (Triglycerides)

Lipids are composed of fatty acids and glycerol. Triglycerides are the major form of fat stored in the adipocytes of your body. Three fatty acids and glycerol combine to form a triglyceride, the fatty acids can be saturated or unsaturated. Glycerol reacts with 3 fatty acids to form a triglyceride.



Properties of fats and oils:

Fats are solids or semi solids while Oils are liquids. Melting points and boiling points are not usually sharp (most fats/oils are mixtures)

Pure fats and oils are colorless and odorless (color and odor is always a result of contaminants) Oils tend to emulsify when shaken with water.

Biological functions of fats and oils:

Fat provides 9 calories per gram, more than twice the number provided by carbohydrates or protein.

Fat is essential for the proper functioning of the body.

Fats provide the "essential" fatty acids. Linoleic acid is the most important essential fatty acid, especially for the growth and development of infants.

Fatty acids provide the raw materials that help in the control of blood pressure, blood clotting, inflammation, and other body functions.

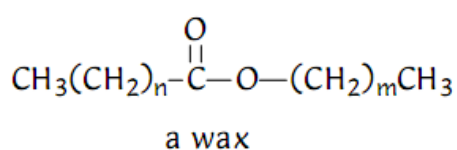
Fat serves as the storage substance for the body's extra calories. It fills the adipose tissue that insulates the body. Fats are also an important energy source.

Healthy skin and hair are maintained by fats.

Fat helps in the absorption, and transport through the bloodstream of the fat-soluble vitamins A,D,E,and K.

 Waxes

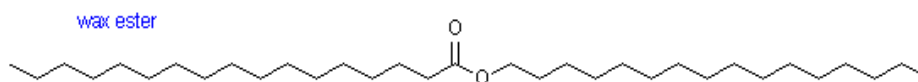
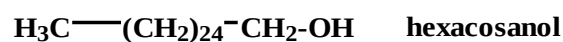
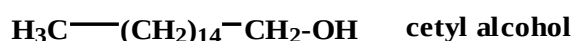
Esters of long-chain fatty alcohols with long-chain fatty acids.



Typically:

$n = 14$ to 34 (even)

$m = 23$ to 35 (odd)

**Examples of long chain monohydric alcohols found in waxes**

-Bees wax: It contains esters derived from alcohols having 24-30 carbon atoms example palmitate of myricyl alcohol and n-hexacosanol. Bees wax was detected in the wall pictures in Egyptian mummies. Ancient Egyptians used it for its adhesive and coating properties, and in ship building. In the Roman period, beeswax was used as a waterproofing agent and treatment for painted walls. Recently beeswax is used as a modeling material, as component of seals, coatings, polishes, and candles.

CH₃ (CH₂)₁₄COOC₃₀H₆₁ myricyl palmitate

CH₃ (CH₂)₁₄COOC₂₆H₅₃ n-hexacosanol palmitate

-Spermacetic wax: It is found in the head cavities and the blubber of the sperm whale. Fatty esters are formed of cetyl palmitate. Spermaceti was used in medicine and later in cosmetics, pharmacy and in candles.

CH₃ (CH₂)₁₄COOC₁₆H₃₃ Cetyl palmitate

-Carnauba wax: Harvested from the dried leaves of carnauba palm, contains wax esters, and a small amounts of free acids and alcohols, hydrocarbons and resins. Carnauba wax consists mainly of fatty acids esterified with tetracosanol and tetratriacontanol alcohols. The wax is used as a floor and automobile wax and as a coating of carbon paper and in the manufacture of polishes.

-Lanolin or wool fat: The grease obtained from the wool of sheep during the cleaning is rich in wax esters. Wool lanolin is a mixture of fatty acid esters of lanosterol and ergosterol. It is used in medical application as a base for creams, ointments and salves. Lanolin has the property of taking up much water without dissolving, which makes it valuable as a medium in the preparation of ointments and cosmetics.

-Insect waxes: The surface of insects is covered by a wax layer, which restrict movement of water across the cuticle and prevent desiccation.

-Microbial waxes: Waxes are not common in prokaryotes, but the mycobacteria produce waxes, termed 'mycoserosates', which is a major virulence factors of pathogenic *Mycobacterium tuberculosis*.

Physiological importance of waxes:

The most important physiological function of waxes is a protective agent on the surfaces of animals and plants.

Waxes are found on the surface of feathers and hair to keep these surfaces soft and pliable.

Waxes prevent aquatic animals from becoming wet.

Waxy coating on the leaves of plants protects them from excessive loss of moisture.

Waxy coating prevents the infection of plants from fungi and bacteria which can cause diseases.

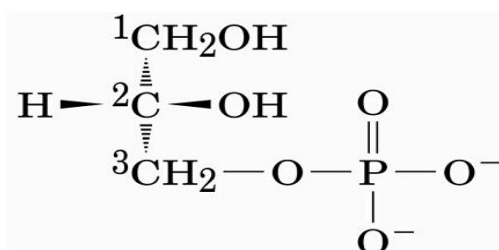
Waxy coating limits the diffusion of water and solutes, permitting a controlled release of volatiles help several fruits like apples and citrus from drying out.

Waxy coating for insects or the skin of animals for protection.

Complex Lipids

❖ Phospholipids (PL):

Contain an ester link between phosphoric acid and an alcohol and also ester link between fatty acid with the same alcohol. The alcohol is either a glycerol to give a glycerophospholipid or a sphingosine to give sphingomyelins.



Phospholipids functions:

Structure Role: predominant lipids of cell membrane.

Role in enzyme action: certain enzyme requires phospholipid for their action.

Role in blood coagulation: required at the stage of conversion of prothrombin.

Role in lipid absorption in intestine: emulsification of water lipid mixture.

Role in electron transport: absolute catalytic requirement of phospholipid for cytochrome oxidase activity.

Role in hormone action: Reservoirs for intracellular messengers signaling after hormone receptor interaction.

Anchor proteins to cell membrane.

Components of lung surfactant & bile (aid solubilization of cholesterol)

Phospholipids classes:

Glycerophospholipids

The most common class of phospholipid containing glycerol alcohol portions.

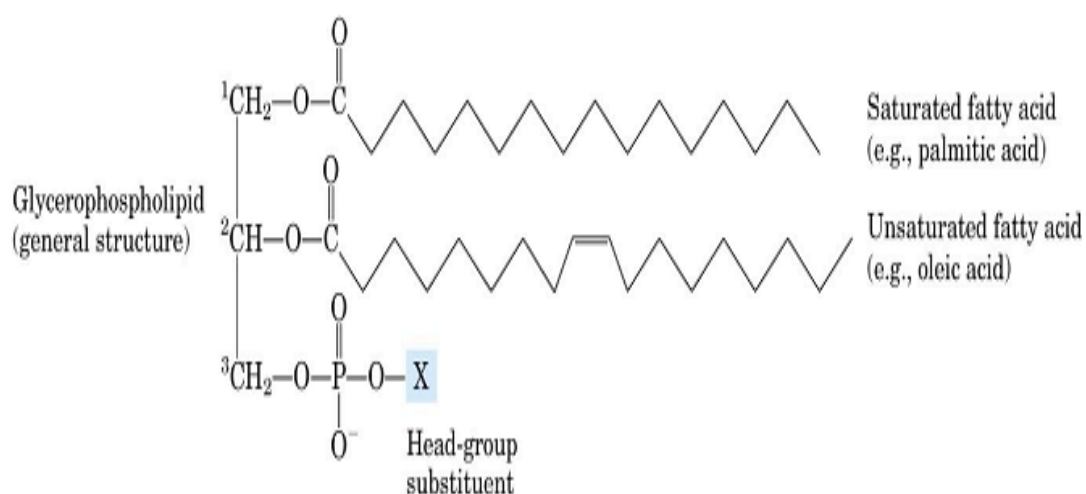
Sphingophospholipids, other class which have non-glycerol alcohol portions sphingosine.

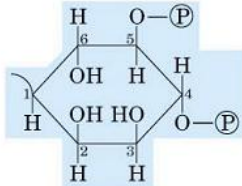
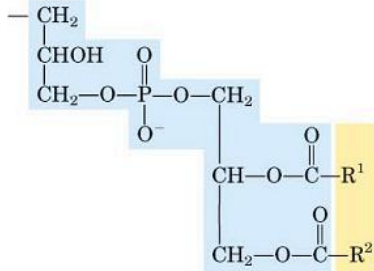
Glycerophospholipids:

Glycerophospholipids are the major component of biological membranes. As with triacylglycerols, glycerophospholipids are composed of a glycerol backbone forming ester linkages to fatty acids. However, the third position is a phosphate ester, rather than a fatty acyl ester. This makes the glycerophospholipid molecule considerably more polar group because it is charged under physiological conditions.

Glycerophospholipids structure:

Phosphatidic acid act as a precursor for some glycerophospholipids. Some common membrane glycerophospholipids are shown below. Note that each is based on the phosphatidic acid structure, with an additional compound esterified to the phosphate. The phosphate and this additional compound are collectively called the head group of the phospholipid. Note that in each case, the head group is a polar molecule.



Name of glycerophospholipid	Name of X	Formula of X	Net charge (at pH 7)
Phosphatidic acid	—	— H	−1
Phosphatidylethanolamine	Ethanolamine	— CH ₂ —CH ₂ —NH ₃ ⁺	0
Phosphatidylcholine	Choline	— CH ₂ —CH ₂ —N ⁺ (CH ₃) ₃	0
Phosphatidylserine	Serine	— CH ₂ —CH(NH ₃ ⁺)—COO [−]	−1
Phosphatidylglycerol	Glycerol	— CH ₂ —CH(OH)—CH ₂ —OH	−1
Phosphatidylinositol 4,5-bisphosphate	<i>myo</i> -Inositol 4,5-bisphosphate		−4
Cardiolipin	Phosphatidylglycerol		−2

Classification of Glycerophospholipids:

Can be classified into several sub-groups according to their molecular structure. All are derived from phosphatidic acid .

Phosphoglycerides containing **one nitrogenous base** such as ethanolamine, choline, serine, and glycine.

A. Choline glycerophospholipids (lecithin):

Phosphatidylcholine is abundant in all cell extracts specially in brain and nervous tissue, it frequently forms about half of all membrane phospholipids in animal preparations. In animal cells, the fatty acid from the 1-position is frequently 16:0 and the 2-position is 18:1 or 18:2, exceptionally more unsaturated fatty acids are found.

B. Phosphatidylethanolamine (cephalin) and phosphatidyl serine:

This phospholipid occurs in nature but usually in concentrations less than 10% of the cell phospholipid pool. In brain tissue, myelin (white matter) has the highest amounts. Phosphatidylserine has generally highly unsaturated acyl chains.

Phosphoglycerides containing **two glycerol molecules** such as diphosphatidylglycerol (cardiolipin) and Phosphatidylglycerol.

C. Phosphatidylglycerol:

It occurs widely but very low amounts are found in animal tissues, in plants it forms 20 to 30 % of total phospholipids (mainly in the chloroplast).

D. Diphosphatidylglycerol:

It is known as **cardiolipin** and is found almost exclusively in mitochondria and in bacteria. Cardiolipin was first discovered in beef heart tissue but later was recognized to be not specific to the heart. Curiously, mammalian cardiolipin contains up to 90 % linoleic acid. This phospholipid has been extensively studied since it was found to be associated to cytochrome oxidase in the electron transport system located in the mitochondrial Membranes, in chloroplast thylakoid membranes, and in bacterial membranes. Cardiolipin seems to be a potential factor in thyroid disease.

Phosphoglycerides containing **inositol** such as phosphoinositides

E. Inositol glycerophospholipids:

Inositol-containing lipids have been reported as early as 1930 in *Tuberculosis bacilli* and later in soybean lipids. The existence of an inositol containing phospholipid could be detected in the animal brain.

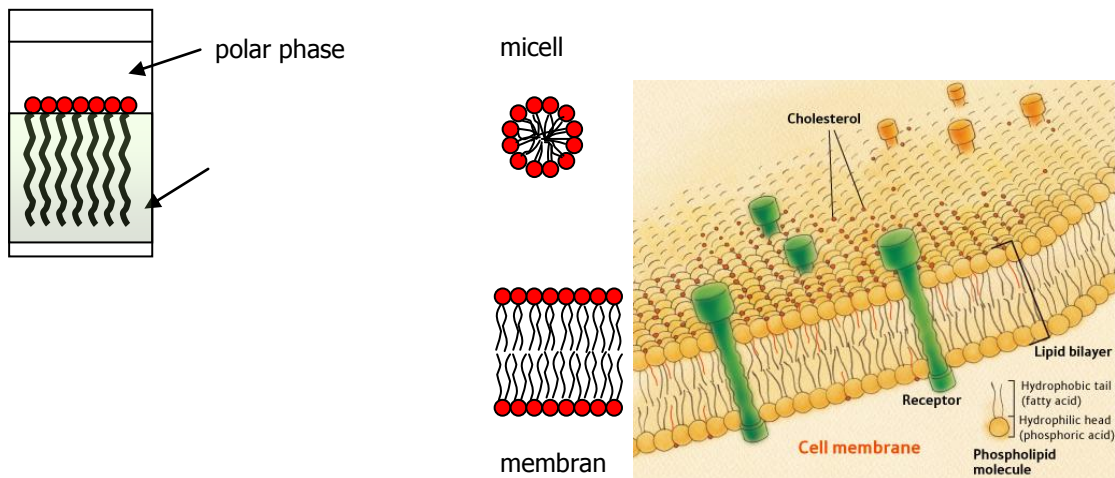
Lysophosphatidic acid:

Formed by removing one fatty acid from phosphatidic acid. This group is a negatively-charged lipid and is the metabolic precursors of all the other similar molecules. Lysophosphatidic acid is abundant in vertebrate serum and seems to have several biological properties including effects on stimulation of DNA synthesis and cell division, calcium regulation, and prevention of programmed cell death, differentiation, and survival. As this compound is liberated during platelet aggregation, it is presumed to play a role in wound healing.

 Cell Membranes:

Phospholipids have a very important structural role in the formation of membranes for cells. They are major components of cell membranes forming lipid bilayers due to amphipathic properties which is essential for phospholipids. Amphipathic molecules

contain an unpolar hydrophobic component (fatty acid) and a highly charged polar head group and hydrophilic (glycerol, phosphate, cholin, colamin, serine, sugar).



Cell Membrane properties:

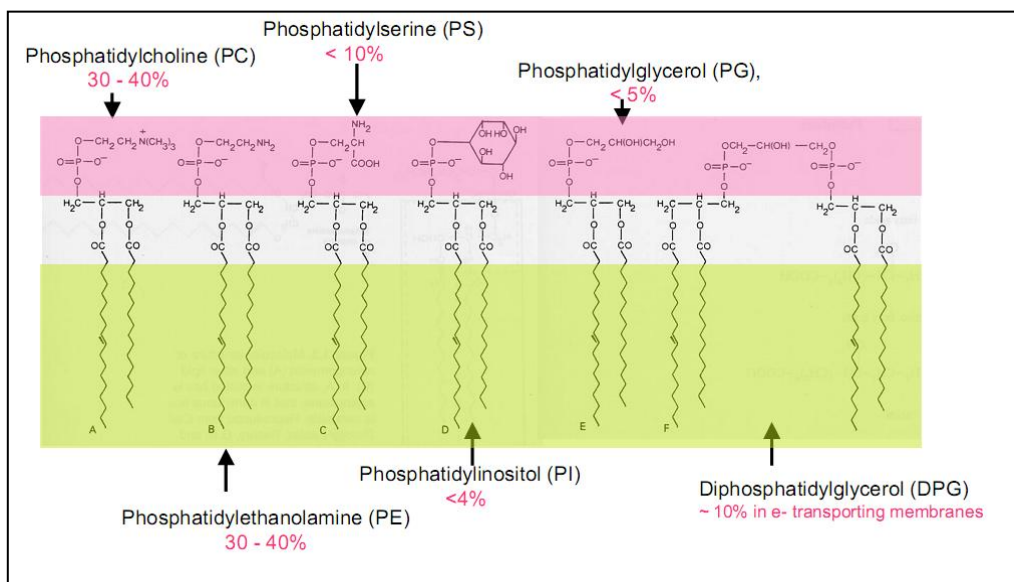
Cell membranes are composed of a fluid like phospholipid bilayer.

Cell membranes control the flow of ions & molecules in & out of cells.

The bilayer incorporates cholesterol, proteins and glycolipids.

cis double bonds in the unsat'd FA residues keep the membranes fluid & flexible.

Small nonpolar molecules cross by diffusion through the lipid bilayer.

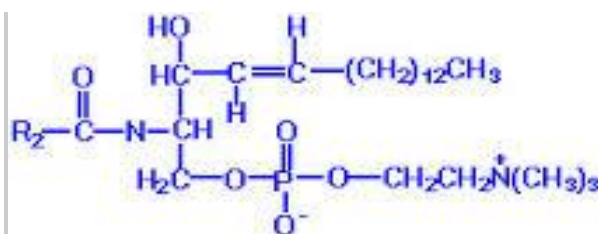


❖ Sphingophospholipids

Sphingolipids, like phospholipids, are composed of a polar head group and non-polar tails. The core of sphingolipids is the long chain-chain amino alcohol, sphingosine.

Sphingomyelins are the only sphingolipid that are phospholipids. They are important

structural lipid components of nerve cell membranes. The predominant sphingomyelins contain palmitic or stearic acid N-acylated at carbon 2 of sphingosine. The sphingomyelins are synthesized by the transfer of phosphorylcholine from phosphatidylcholine to a ceramide in a reaction catalyzed by sphingomyelin synthase.

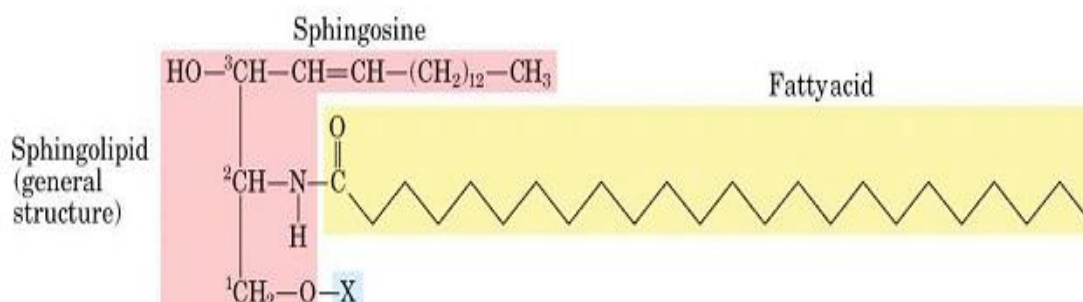


❖ Glycolipids

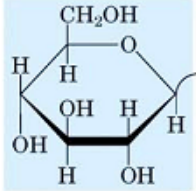
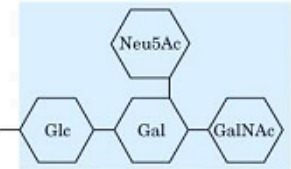
Glycolipids are carbohydrate-attached lipids. Their role is to provide energy and also serve as markers for cellular recognition. They occur where a carbohydrate chain is associated with phospholipids on the exoplasmic surface of the cell membrane. The carbohydrates are found on the outer surface of all eukaryotic cell membranes. They extend from the phospholipid bilayer into the aqueous environment outside the cell where it acts as a recognition site for specific chemicals as well as helping to maintain the stability of the membrane and attaching cells to one another to form tissues.

Types of glycolipid:

Sphingolipids like phospholipids, are composed of a polar head group and non-polar tails. The core of sphingolipids is the long chain amino alcohol, sphingosine. Amino acylation, with a long chain fatty acid, at carbon 2 of sphingosine yields ceramide the precursor for some glycolipids. Glycosphingolipids, or glycolipids, are composed of a ceramide backbone with a wide variety of carbohydrate groups (mono- or oligosaccharides) attached to carbon 1 of sphingosine. The four principal classes of



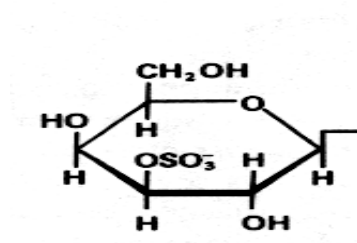
glycosphingolipids are the cerebroside, sulfatides, globosides and gangliosides.

Name of sphingolipid	Name of X	Formula of X
Ceramide	—	— H
Sphingomyelin	Phosphocholine	$\begin{array}{c} \text{O} \\ \parallel \\ -\text{P}-\text{O}-\text{CH}_2-\text{CH}_2-\text{N}^+(\text{CH}_3)_3 \\ \\ \text{O}^- \end{array}$
Neutral glycolipids Glucosylcerebroside	Glucose	
Lactosylceramide (a globoside)	Di-, tri-, or tetrasaccharide	
Ganglioside GM2	Complex oligosaccharide	

❖ Cerebrosides:

Cerebrosides are sphingolipids made by attaching a sugar to a ceramide. The most common cerebrosides include galactocerebrosides, glucocerebrosides, and Galactocerebrosides are found predominantly in neuronal cell membranes. Glucocerebrosides are not found in membranes, instead, they represent intermediates in the synthesis or degradation of more complex glycosphingolipids.

Sulfatides: The sulfuric acid esters of galactocerebrosides.



Gangliosides:

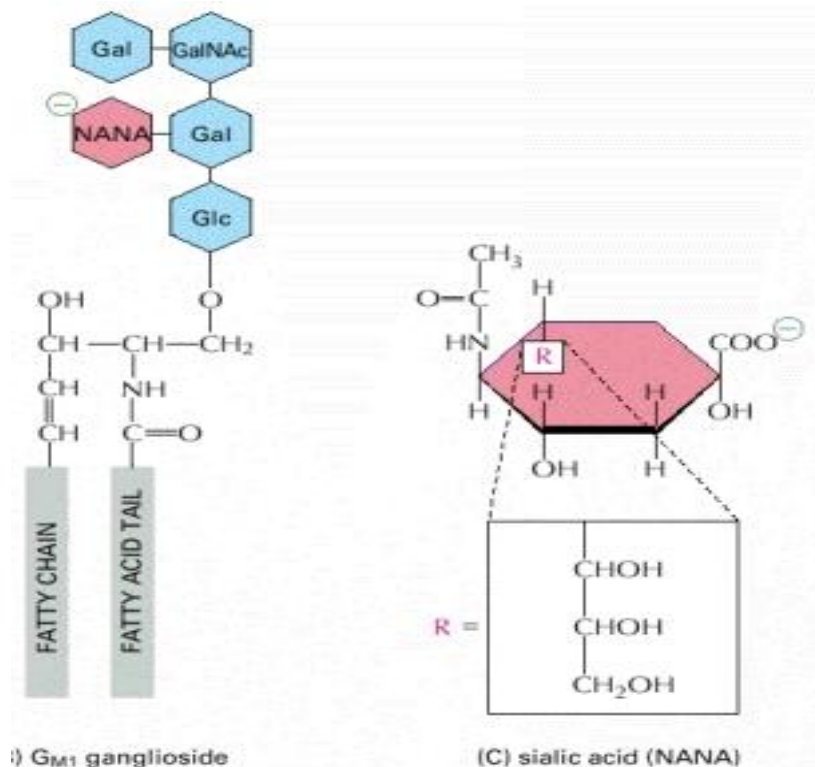
Gangliosides differ from cerebrosides in containing sialic acid. This group includes molecules composed of ceramide linked by a glycosidic bond to an oligosaccharide chain containing hexose and N-acetylneuraminic acid (NANA), which is acidic sugar

known also as sialic acid. Gangliosides are characterized by a high amount of stearic acid (C18, about 80%), the rest being C16, C20 and C22.

✚ Nomenclature:

letter G refers to the name ganglioside

the subscripts M, D, T and Q indicate mono-, di-, tri-, and quatra(tetra)-sialic-containing gangliosides. The numerical subscripts 1, 2, and 3 refers to the order of migration of the ganglioside on thin-layer chromatography (e.g., GM3 > GM2 > GM1).

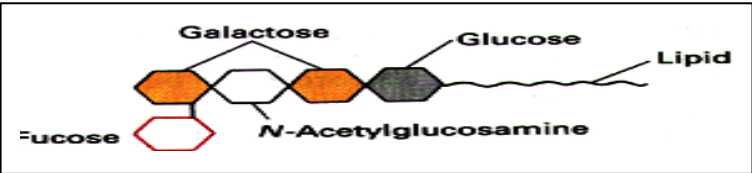
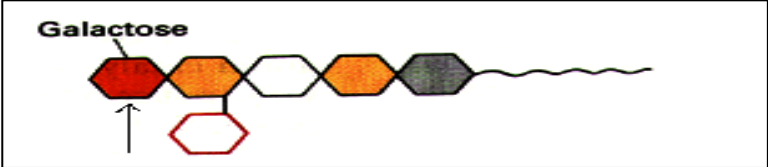
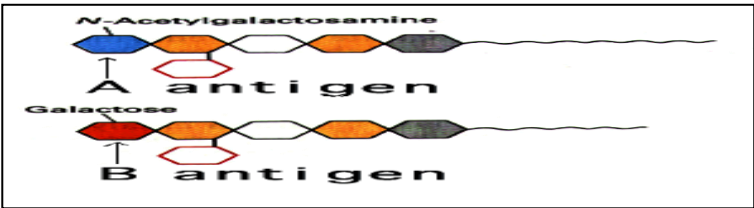


In the case of abnormal conditions in which genetic defects in catabolism occur leads to ganglioside accumulations. GM1 accumulates in the nervous system leading to mental retardation and liver enlargement. In the Tay-Sachs syndrome GM2 accumulates in brain tissue leading to mental retardation and blindness.

✚ Blood group Antigens

One of the glycolipids found in human red blood cells is involved in the antigens of the ABO blood type. Some glycoproteins are also involved with the antigens as well. The antigens which determine blood types belong to glycoproteins and glycolipids. There are three types of blood-group antigens: O, A, and B. They differ only slightly in the composition of carbohydrates. Humans contain enzymes which catalyze O

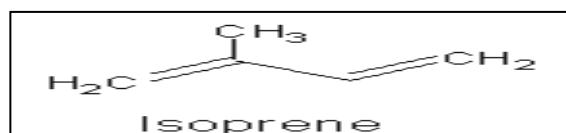
antigen synthesis. Humans with A-type blood also contain an additional enzyme which adds N-Acetylgalactosamine to the O antigen. Humans with B-type blood contain another enzyme (called B-type enzyme) which adds Galactose to the O antigen. Humans with AB-type blood contain both A-type and B-type enzymes while humans with -type O blood lack both types of enzymes.

blood	Glycolipid	Antibodies in blood
O	 RBC of O type blood have this glycolipids in their RBCs plasma membranes.	anti A and anti B Individuals with type O blood have antibodies against the type A and type B antigens.
A	 The A type glycolipid has the same carbohydrate composition as does the type O antigen with the addition of a additional N-Acetylgalactosamine which is the A type antigenic determinant.	anti B individuals of type A blood have antibodies against the type B determinant
B	 The B type glycolipid has the same carbohydrate composition as does the type O antigen with the addition of an additional galactose which is the B type antigenic determinant.	anti A Individuals of type B blood have antibodies against the type A antigenic determinan
AB	 Individuals with type AB blood have both of the above glycolipids in their RBC plasma membranes	neither A or B Individuals with their AB blood have antibodies against neither the the A nor B antigens

Precursor and derived lipids

Terpenes

A large family of plant compounds that play a variety of roles in many different plants and are constructed from isoprenoid units. The isoprene contains five carbons. Isoprene units are linked **head-to-tail**.



Classification:

Terpenes classification depends upon the number of isoprene units incorporated in the basic molecular skeleton. Monoterpene (10C) (two units), Sesquiterpene (15C) (three units), Diterpene (20C) (four units), Triterpene (30C) (six units), and Tetraterpenes (40C) (eight units).

Function:

Many terpenes play roles as plant hormones and in the chemical defenses of plants against microbial diseases and insect; many others have important medical properties. Terpenes and terpenoids are the most important constituents in essential oils.

Terpene and steroid biosynthetic paths.

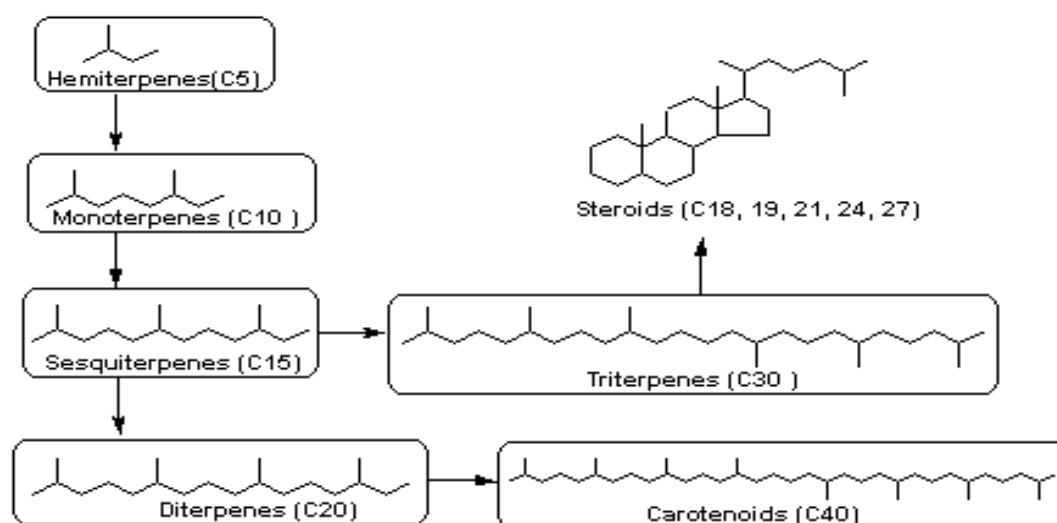
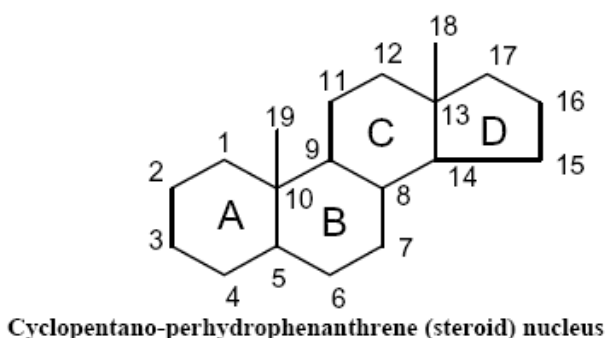


Fig.1 Biosynthetic Pathway of Terpene Skeletons

✚ Steroids:

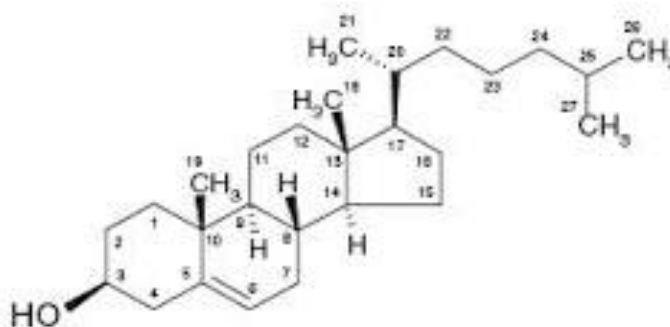
Over 30 steroids found in plant, animal tissues, and yeasts but not bacteria, and may exist free or combined with fatty acids or carbohydrates. All of the steroids have a similar cyclic nucleus resembling phenanthrene (rings A, B, and C) to which acyclopentane ring (D) is attached. Several steroids contain, one, two or three double bonds and many possess one or more hydroxyl groups. Most steroids contain methyl groups at carbon 10 and 13 and a side chain at carbon 17. They function as hormones and other regulators of physiological processes.



❖ Cholesterol:

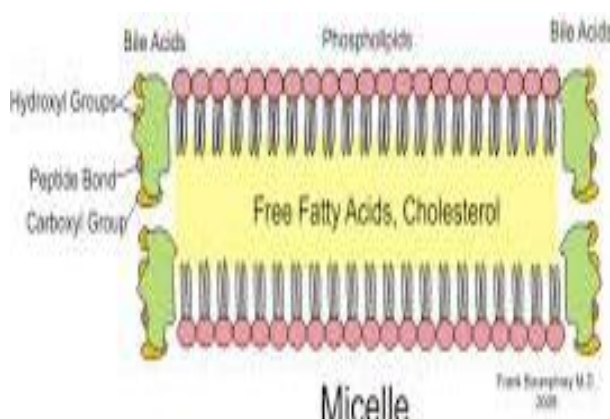
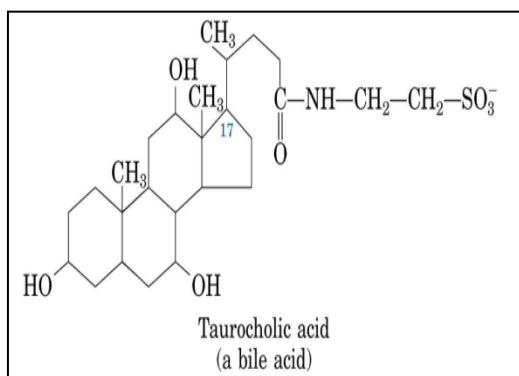
It is the best known and most abundant steroid in the human body (about 240 gm.), found in cellular membranes, the brain and nerve tissue. Most meats, eggs, butter, and icecream are rich in cholesterol. Cholesterol is synthesized in the liver from acetyl coenzyme A. The human body synthesizes about 3 gm cholesterol each day. If the precursor of cholesterol is present in the diet, then its biosynthesis in the liver is suppressed. There are two types of blood cholesterol. Low density lipoprotein (LDL) cholesterol and high density lipoprotein (HDL) cholesterol. LDL is considered the 'bad' cholesterol because it contributes to the narrowing and silting up of the arteries, which can lead to heart disease and stroke. The HDL cholesterol is considered to be the 'good' cholesterol because it actually cleans out the arteries, reducing the risk of cardiovascular disease.

Cholesterol content of blood varies with age, diet, and sex. Females tend to have lower blood cholesterol levels than males. When cholesterol precipitates out of the blood and accumulates in the blood vessels, the resulting constriction reduces blood flow and leads to high blood pressure. Reducing blood cholesterol level is by minimizing lipid intake through dietary restriction, or by administering anticholesterol drugs, will help to prevent heart disease.



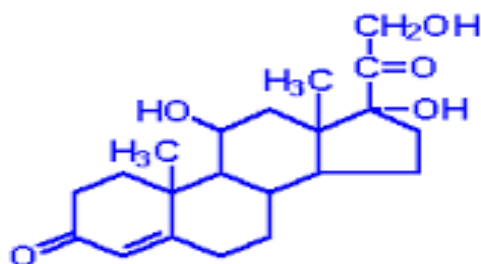
❖ Bile Salts:

Are sodium salts of bile acids and amino acids, synthesized from cholesterol in the liver, stored in the gall bladder, and secreted into the intestine. Bile salts are bile acids compounded with a [cation](#), usually [sodium](#). In humans, the salts of [taurocholic acid](#) and [glycocholic acid](#) derivatives of [cholic acid](#) represent approximately eighty percent of all bile salts. They aid in the breakdown and absorption of dietary lipids. They are highly effective detergents because they contain both hydrophobic and hydrophilic groups. They act as emulsifying agents, break down fat globules into smaller ones and keep these smaller globules suspended in the aqueous digestive environment.



❖ Adrenocortical Hormones:

The outer part or cortex of the adrenal glands utilizes cholesterol to produce a mixture of many steroids essential to life. The steroids constitute hormones as cortisone can be considered to be the parent which effect the metabolism of foodstuffs, maintain the proper balance of electrolytes as sodium and potassium ions and control inflammation.



❖ VitaminD:

A closely related group of compounds that are effective in preventing rickets and include vitamin D₂ (ergocalciferol) and vitamin D₃(cholecalciferol). Both vitamins are produced by ultraviolet irradiation of unsaturated steroids.

Special Features of Lipids

Rancidity

The term rancid is applied to **any fat or oil that develops a disagreeable odor**. Two principal chemical reactions are responsible for causing rancidity -**hydrolysis** and **oxidation**.

Butter is particularly susceptible to hydrolytic rancidity because it contains many of the lower molar mass acids (butyric, caproic) all of which have offensive odors. Under moist and warm conditions, hydrolysis of the ester linkages occurs, liberating the volatile acids. Microorganisms present in the air furnish the enzymes (lipases) that catalyze the process. Storing butter covered in a refrigerator can easily prevent rancidity.

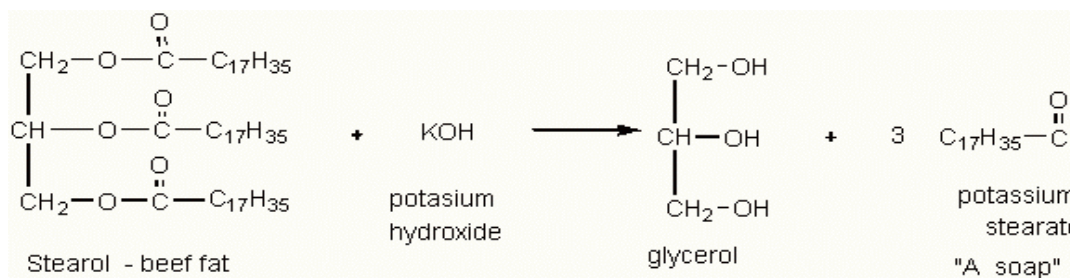
Oxidative rancidity occurs in triacylglycerols containing polyunsaturated fatty acids. The reaction is quite complex, but it is believed that the first step involves the formation of a free radical, followed by production of hydroperoxides. Further oxidation reactions occur in which bonds are cleaved and the short-chain, offensive -smelling carboxylic acids are liberated.

Rancidity is a major concern of the food industry, and chemists involved in this area are continually seeking new and better substances to act as **Antioxidants**. Such compounds are added in very small amounts (0.001-0.1 %) to suppress rancidity. They have a greater affinity for oxygen than the lipid to which they are added and thus function by preferentially depleting the supply of adsorbed oxygen. Two of the naturally occurring antioxidants are tocopherol (vitamin E) and ascorbic acid (vitamin C).

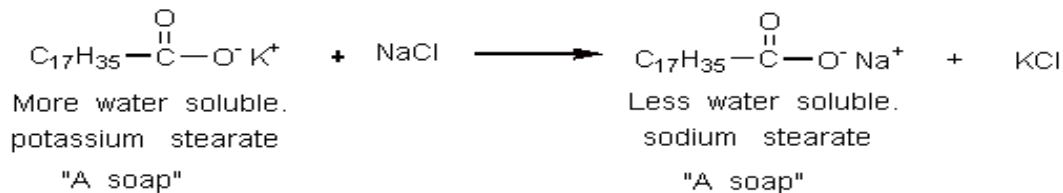
Soaps, Detergents and Cleaning

❖ Soaps

Fats are isolated from plants and animals. The properties such as solubility relate to their chemical structures. Fats are heated with a strong base to convert them into soaps. The fat you use to make soap reacts with potassium hydroxide to produce potassium soap, the potassium salt of the fatty acid. One typical animal fat, sterol, reacts with KOH to form potassium stearate, soap. Most naturally occurring fats produce a mixture of different salts of fatty acids when they are converted to soap.



The potassium soap formed from your fat is converted to sodium soap by replacing the potassium ions with sodium ions. A large excess of sodium chloride supplies the sodium ion. You may also notice that the potassium soap is softer than the sodium soap. In addition there is a difference in the way the sodium and potassium soaps behave in water.



The nonpolar ends of the molecule associate with the fat, grime or dirt which is also nonpolar, The polar or ionic end of the molecule attracts the water molecules. A spherical structure with the polar portions of the molecule on the surface and the nonpolar parts of the molecule in the center is attracted to the water and carries the non-water-soluble material away with it. This spherical shaped unit of soap and grime is a micelle.

Magnesium and calcium salts of the same fatty acids that make up potassium and sodium soaps are not water soluble. When sodium or potassium soaps are put into water containing calcium and magnesium ions, the cloudiness, scum or curds consist of less soluble calcium and magnesium soaps. To achieve the same washing or cleaning action, more soap must be added.

❖ Detergents:

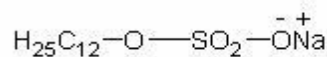
A detergent is a non-soapy cleaning agent that uses a surface-active agent for cleaning a substance in solution. Unlike soaps they are effective even in hard or salt water, as they form no scum.

Modern synthetic detergents are alkyl or aryl sulphonates produced from petroleum (or coal) and sulphuric acid. They can be defined as 'the sodium or potassium salt of a long chain alkyl

benzene sulphonic acid or the sodium or potassium salt of a long chain alkyl hydrogen sulphate that have cleansing properties in water'. Like soaps, detergents contain one large non-polar hydrocarbon group and one short ionic or highly polar group at each end, which allow for the cleansing action of dirt in water. Two basic examples of known detergents of the sulphonate group($-\text{SO}_3\text{-Na}^+$) or the sulphate group($-\text{OSO}_3\text{-Na}^+$) are:



Sodium p-dodecyl benzene sulphonate

Sodium lauryl sulphate or
Sodium n-dodecyl sulphate

Soaps	Detergents
They are metal salts of long chain higher fatty acids.	These are sodium salts of long chain hydrocarbons like alkyl sulphates or alkyl benzene sulphonates.
These are prepared from vegetable oils and animal fats.	They are prepared from hydrocarbons of petroleum or coal.
They cannot be used effectively in hard water as they produce scum i.e., insoluble precipitates of Ca^{2+} , Mg^{2+} , Fe^{2+} etc.	These do not produce insoluble precipitates in hard water. They are effective in soft, hard or salt water.
These cannot be used in acid solutions.	They can be used even in acid solutions.
Their cleansing action is not as strong as that of detergents.	Their cleansing action is by surfactants, which is a strong cleansing action.
These are biodegradable.	Some of these are not biodegradable.

If a straight chain hydrocarbon is used in the detergent instead of a branched chain hydrocarbon, then the detergent becomes biodegradable. Thus the major disadvantage of detergents can be overcome